# APPLICATION OF MEDICINAL MUSHROOM EXTRACTS IN SHELF-STABLE GLUTEN-FREE PRODUCTS: PHYSICOCHEMICAL, ANTIOXIDANT AND SENSORY EVALUATION


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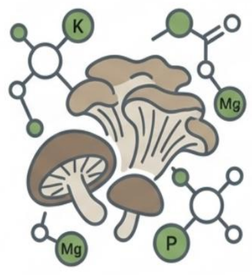
### The Challenge

**Technological & Nutritional Deficits**

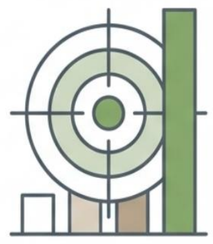
Rice flour provides critical low allergenicity for gluten-free diets, but the absence of gluten causes severe structural deficiencies, high cooking losses, and low baseline nutritional value.




### The Functional Solution

**Medicinal Mushroom Extracts**

Shiitake and Maitake extracts are rich in proteins, minerals (K, Mg, P), beta-glucans, and antioxidants. Mushroom polysaccharides interact with rice starch to improve structural integrity, while polyphenols drastically increase oxidative stability.



### The Primary Goal

**Matrix Evaluation**

To rigorously evaluate the technological, physicochemical, antioxidant, and sensory impact of these specific extracts across two distinct thermal and mechanical matrices: boiled (pasta) and baked (crackers).

## Formulation Matrices

Formulation of gluten-free pasta variants (per 100 g of composite flour)

| Sample | Rice Flour (g) | Shiitake (g) | Maitake (g) | Egg (g) | Tapioca Starch (g) | Water (mL) | Olive Oil (mL) | Salt (g) |
|---|---|---|---|---|---|---|---|---|
| P0 | 100 | - | - | - | 25 | 50 | 2,5 | 0,2 |
| PS15 | 85 | 15 | - | - | 25 | 45 | 2,5 | 0,2 |
| PM5 | 95 | - | 5 | - | 25 | 60 | 2,5 | 0,2 |
| P0-E | 100 | - | - | 25 | - | 20 | - | 0,2 |
| PS15-E | 85 | 15 | - | 25 | - | 5 | - | 0,2 |
| PM5-E | 95 | - | 5 | 25 | - | 10 | - | 0,2 |

**Legend:** P0 – egg-free pasta (control); PS15 – pasta with 15% shiitake extract; PM5 – pasta with 5% maitake extract; P0-E – egg pasta; PS15-E – egg pasta with 15% shiitake extract; PM5-E – egg pasta with 5% maitake extract

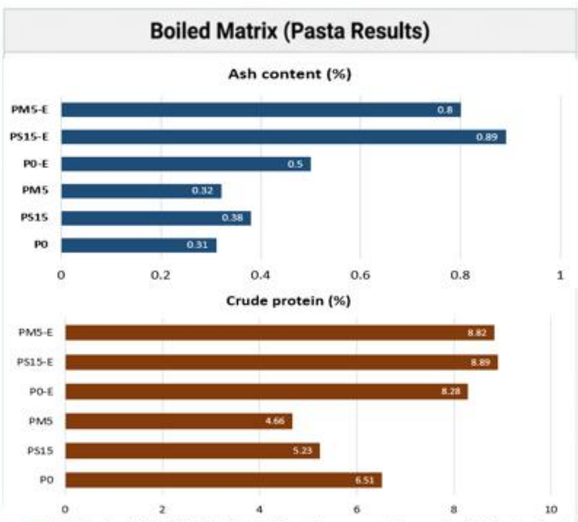
Recipe for the preparation of gluten-free crackers (per 100 g rice flour)

| Simple | Rice Flour (g) | Shiitake (g) | Maitake (g) | Olive Oil (mL) | Vinegar (mL) | Sodium bicarbonate (g) | Water (mL) | Salt (g) |
|---|---|---|---|---|---|---|---|---|
| C | 100 | - | - | 10 | 1,5 | 1 | 60 | 1,5 |
| CS5 | 95 | 5 | - | 10 | 1,5 | 1 | 65 | 1,5 |
| CM5 | 95 | - | 5 | 10 | 1,5 | 1 | 65 | 1,5 |
| CS10 | 90 | 10 | - | 10 | 1,5 | 1 | 60 | 1,5 |
| CM10 | 90 | - | 10 | 10 | 1,5 | 1 | 60 | 1,5 |

**Legend:** C – control crackers; CS5 – crackers with 5% shiitake extract; CM5 – crackers with 5% maitake extract; CS10 – crackers with 10% shiitake extract; CM10 – crackers with 10% maitake extract

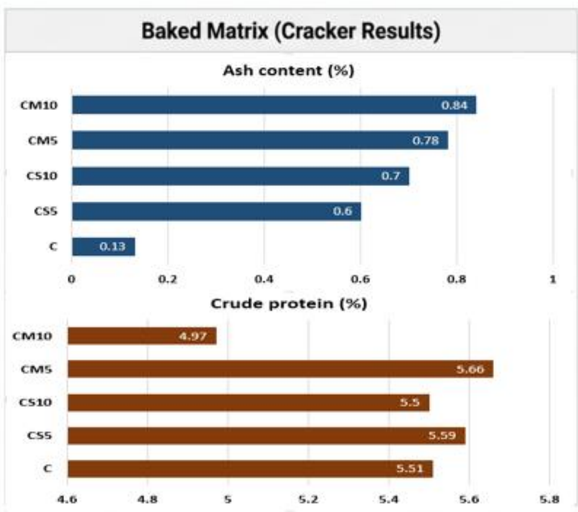
## Mushroom Extracts Enhance Mineral Profile with Slight Protein Variations

Incorporation of extracts significantly alters the foundational chemical composition across both matrices, according to AOAC standards.




**Ash Content (%):** Statistically significant increases observed in fortified variants.

**Crude Protein (%):** Slight relative decreases in crude protein proportion observed in certain non-egg variants compared to the control.

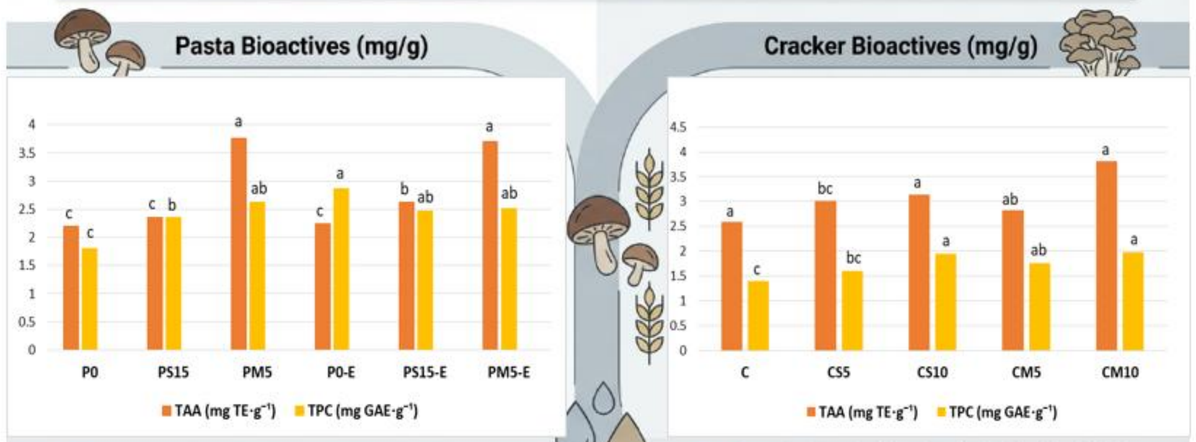



**Ash Content (%):** Pronounced, dose-dependent increase correlating with extract concentration.

**Crude Protein (%):** Mirrors the pasta trend with slight protein adjustments based on the extract substitution ratio.

## Significant Surges in Antioxidant Activity and Phenolic Content

**Insight:** Both Shiitake and Maitake extracts successfully transformed basic gluten-free matrices into functional foods with high bioactive potential (TAA: $p<0.0001$, TPC: $p<0.05$)



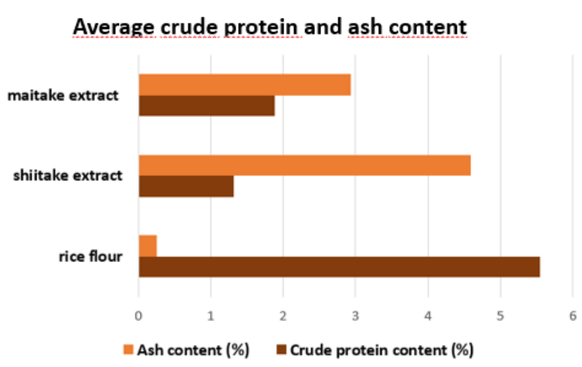

TPC and AA spiked dramatically in PS15, PM5, PS15-E, and PM5-E compared to the baseline P0/P0-E controls.

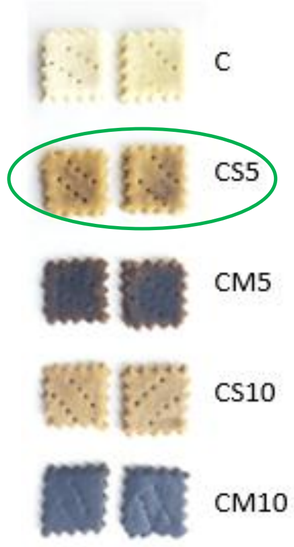
Demonstrates a clear dose-dependent relationship. The 10% extract variants (CS10, CM10) yielded the absolute highest TPC and AA scores.

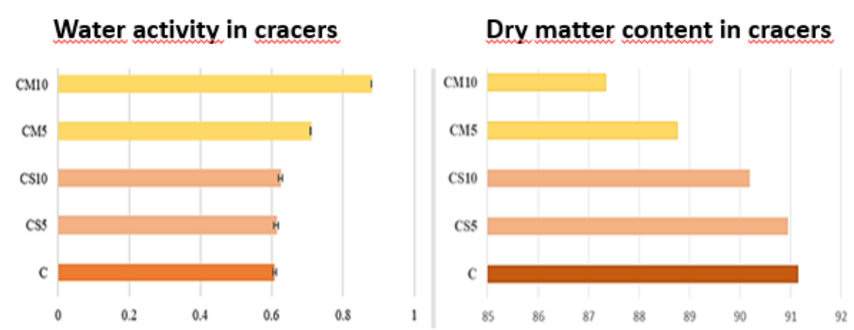
**Legend:** P0 – egg-free pasta (control); PS15 – pasta with 15% shiitake extract; PM5 – pasta with 5% maitake extract; P0-E – egg pasta; PS15-E – egg pasta with 15% shiitake extract; PM5-E – egg pasta with 5% maitake extract. Values are expressed as mean ± standard deviation (n=8). Means in the same row followed by different letters are significantly different (TAA: $p<0.0001$, TPC: $p<0.05$).

**Legend:** C – control crackers; CS5 – crackers with 5% shiitake extract; CM5 – crackers with 5% maitake extract; CS10 – crackers with 10% shiitake extract; CM10 – crackers with 10% maitake extract.

Attractive color profile of samples with the highest sensory ratings

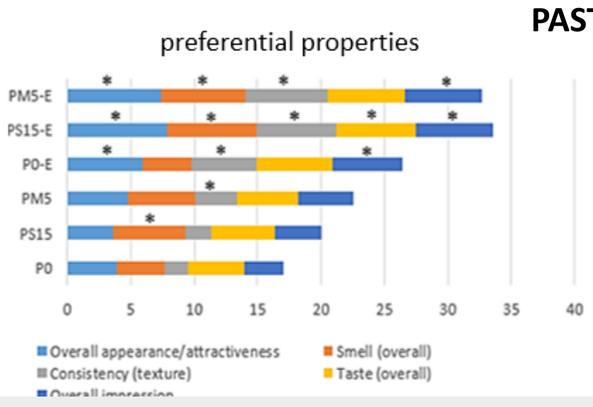


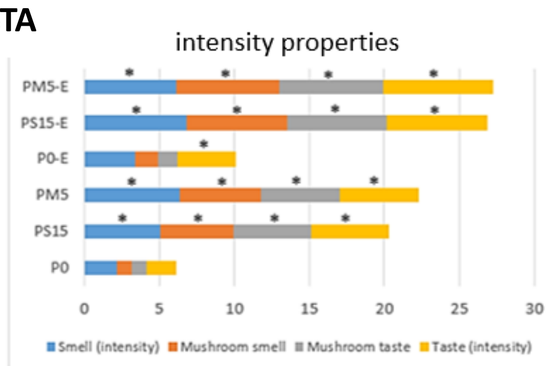



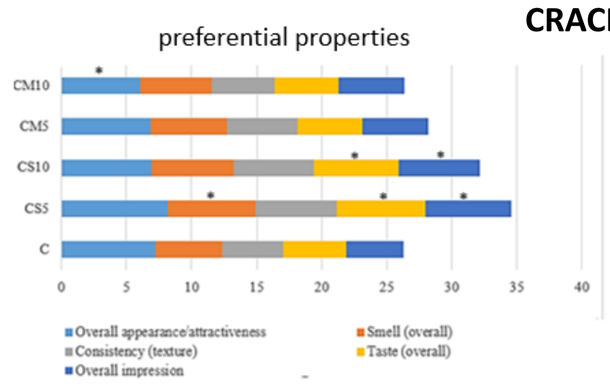



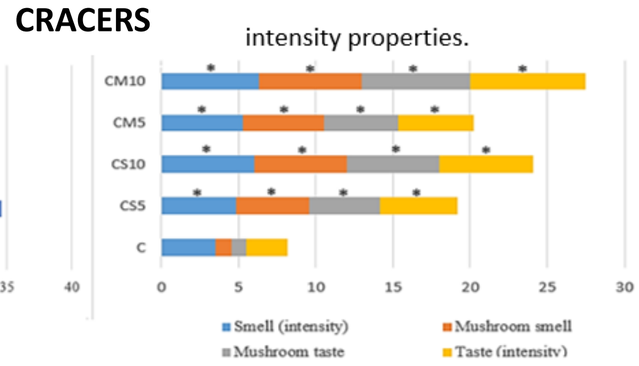

The decreasing trend in dry matter and the simultaneous increase in water activity ($a_w$) with increasing extract concentrations requires adjustment of the composition to ensure microbiological stability.

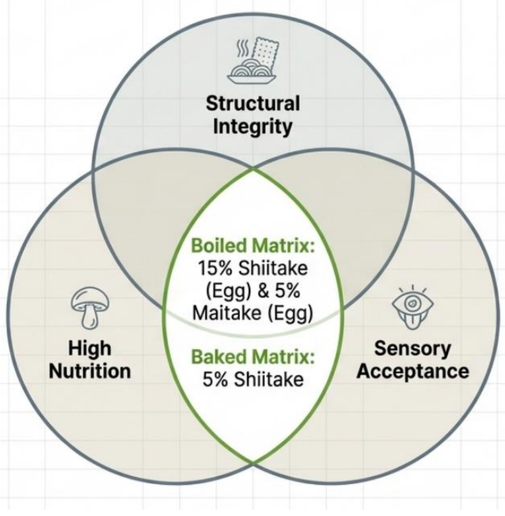
## Sensory evaluation of products with added shiitake and maitake extracts





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## CONCLUSION

This study confirms that the incorporation of Lentinula edodes and Grifola frondosa extracts modifies the nutritional and bioactive profiles of gluten-free pasta and rice-based crackers. The addition of these extracts increased total phenolic content (TPC) and antioxidant activity in both product types.

Among gluten-free pasta, the 15% shiitake and 5% maitake egg-based variants achieved the highest consumer acceptance for appearance and flavour. In the case of rice-based crackers, the 5% shiitake variant was identified as optimal for overall impression. Higher concentrations (10%) led to significant surface darkening, which reduced visual scores despite the increased antioxidant potential.

In conclusion, mushroom extracts are suitable functional ingredients for gluten-free cereal products. Future research should focus on further optimizing the texture of non-egg formulations and managing browning intensity in baked snacks to balance the enrichment of bioactive compounds with sensory requirements.


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